```
install.packages("ggplot2")
library(ggplot2)
install.packages("dplyr")
library(dplyr)
install.packages("broom")
library(broom)
install.packages("ggpubr")
library(ggpubr)

#Import data set
#Check summary of the data
summary(income_data)
#Check whether the dependent variable follows normal distribution
hist(income_data$happiness)

#scatter plot
ggplot(income_data,aes(x=income,y=happiness))+geom_point()

ggplot(income_data,aes(x=income,y=happiness))+geom_line()

#Regression line
ggplot(income_data,aes(x=income,y=happiness))+geom_point() +
  geom_smooth(method='lm')

#code for linear model
income.happiness.lm <- lm(happiness ~ income, data = income_data)
#summary of the model
summary(income.happiness.lm)

#plot regression line
cor(income_data)
ggplot(income_data, aes(x=income, y=happiness))+geom_point()+
  geom_smooth(method="lm", col="red")+
  stat_regline_equation()+
  labs(title = "Income Vs Happiness")

#OR
income.graph=ggplot(income_data, aes(x=income, y=happiness))+geom_point()


income.graph=income.graph + geom_smooth(method="lm", col="red")
income.graph

#Equation of regression line
income.graph <- income.graph +
  stat_regline_equation()
income.graph

#title assign to graph
income.graph +
  theme_transparent() +
```

```
  labs(title = "Reported happiness as a function of income",
       x = "Income (x$10,000)",
       y = "Happiness score (0 to 10)")
```